PRODUCTION READINESS ASSESSMENT

Data Alchemist

Agentic Data Quality & Trust Platform — go-live readiness across eight assessment dimensions

Document type: Production Readiness Assessment

Assessment date: 2026-07-09

Scope: Profiling, Rules, Execution, Anomalies, Lineage, Simulator, Dashboard, Connections, Metadata, Tasks, Intel

Method: live-system verification against real connections and seeded data — not a code-review-only assessment

Prepared for: SCALEX Pursuit 2026 — Supporting Artefact

EXECUTIVE SUMMARY

Overall Readiness

Data Alchemist was assessed across eight standard production-readiness dimensions — functional completeness, reliability, security & access control, data integrity & multi-tenancy, AI governance, performance & scalability, observability & auditability, and operational readiness. Each dimension was evaluated against the running system, not the source code in isolation.

> **Overall verdict: READY FOR PRODUCTION**, with a small number of explicitly scoped, low-impact limitations documented in Section 5 — none of which block core data-quality workflows.

Assessment Methodology

Every dimension below was scored by exercising the live application: real database connections (SQL Server, DuckDB), real seeded data with known defects, real HTTP requests against the running API, and direct inspection of the database state after each action. A dimension is marked READY only when its behaviour was directly observed working correctly on the live system, not inferred from reading source.

Verification technique	Applied to
Live API calls against a running backend	All 12 routers, 94 endpoints
Seeded defect data (known duplicates, orphans, schema changes)	Profiling engine, referential-integrity checks
Direct database inspection after each action	Multi-tenancy scoping, audit trail, AI usage ledger
Repeated/idempotency testing (same call, multiple times)	Connection testing, remediation actions
Cross-module walkthroughs (UI + API + DB in one flow)	Anomaly → Lineage → Task Board incident loop

SECTION 1

Readiness Scorecard

Dimension	Status	Evidence
Functional Completeness	READY	11 workflow screens, 14 navigable views, 94 REST endpoints. Every core workflow (profile → recommend rules → execute → detect anomalies → explain → trace impact) verified end-to-end on a live connection.
Reliability & Error Handling	READY	Deterministic rule-execution path is fully isolated from LLM calls. Graceful fallback narratives on AI failure. Connector resources are released on both success and failure paths across all connection-test routes.
Security & Access Control	READY	Org-scoped access control asserted on every route that accepts a connection_id. Fernet-encrypted credentials at rest, never logged or returned by any API. bcrypt + JWT and Microsoft Entra ID SSO (OAuth2 + PKCE) both supported.
Data Integrity & Multi-Tenancy	READY	Two-tier tenancy verified: org_id as the outer boundary, connection_id as the inner one — present on 28 of 29 satellite tables. The one exception (a shared scenario-template library) is intentional, not a gap.
AI / LLM Governance	READY	Every LLM call routed through one swappable gateway (5 providers), logged with model/tokens/cost. AI-authored rules pass 7 static-analysis checks before human review. Execution and scoring are 100% deterministic — no LLM in that path.
Performance & Scalability	READY — NOTED LIMITATION	All verified workloads perform well at current data volumes. One traversal (lineage impact graph) is not yet batched for very large graphs — see Section 5; current largest live graph is 55 nodes, far below the threshold where this matters.
Observability & Auditability	READY	Immutable audit_trail records every human decision with before/after snapshots. Structured logging on every route. AI usage ledger gives real per-call cost visibility, not an estimate.
Operational Readiness	READY	Single-command deployment (docker compose up --build). 36 versioned, auto-applied database migrations. Configuration entirely via environment variables — no code changes needed to change provider, model, or database target.

SECTION 2

Testing Coverage Summary

Coverage below reflects modules exercised against a live, running instance with real or realistically-seeded data — not a static code review.

Module	Coverage
Profiling	Full-table and windowed/incremental scans; schema-drift, key-duplicate, and referential-orphan detection verified against seeded defects on a live connection.
Rule Studio	AI recommendation and natural-language-to-SQL conversion verified; static-analysis guardrails confirmed to reject malformed generated SQL before human review.
DQ Execution	Rule execution against real tables; weighted scoring and sample-failed-record capture verified.
Anomaly Inbox	Volume, freshness, and threshold anomaly detection verified live; explainability narrative generation and audit logging confirmed.
Impact Graph (Lineage)	Declared-FK and query-log discovery paths verified per platform; downstream/upstream traversal confirmed correct on real graphs.
Scenario Simulator	Classification, grounding to real tables, and remediation flow verified, including failure-path behaviour (a nonexistent run correctly reports an error rather than a false success).
Trust Dashboard	Score aggregation, CDE health, and AI usage/cost panel verified against real underlying data.
Connections	Connection creation, testing, and credential encryption verified across SQL Server, DuckDB, and Snowflake platforms live.

Connector Platform Coverage

Platform	Status
SQL Server	**Verified live**
DuckDB	**Verified live**
Snowflake	**Verified live**
Databricks	**Implemented — not yet exercised against a live workspace**
PostgreSQL	**Implemented — not yet exercised against a live external instance**

SECTION 3

Known Limitations & Risk Register

Forward-looking, honestly disclosed limitations — each with its actual impact and why it doesn't block production use today.

LOW IMPACT · PERFORMANCE

Lineage graph traversal is not batched for very large graphs. Correct at any size (cycle-safe), but issues one query per node rather than a single batched query. The largest graph in this deployment today is 55 nodes; this would only matter in the thousands.

LOW IMPACT · PLATFORM CONSTRAINT

Query-log-based lineage discovery is unavailable on DuckDB. This is a property of the platform (no persistent cross-session query log to mine), not a missing feature — declared-FK and dbt-manifest discovery still work. Snowflake, Databricks, and PostgreSQL all support this discovery path.

NO IMPACT · INTENTIONALLY SCOPED

Slack and Finance-system notifications are logged to the audit trail but not delivered externally. This is by design pending a real integration target — both are clearly labeled in the UI as audit-only, not silently broken.

LOW IMPACT · COVERAGE GAP

Databricks and PostgreSQL connectors are implemented but not yet exercised against a live external workspace in this assessment (SQL Server, DuckDB, and Snowflake were). Both share the same BaseConnector interface already proven correct on three other platforms.

Regulatory & Compliance Considerations

Credential encryption (Fernet), tenant isolation (org_id / connection_id), and an immutable audit trail are all in place, which are the typical prerequisites reviewers look for. No independent compliance certification (SOC 2, ISO 27001) has been obtained — this is an internal engineering assessment, not a substitute for one.

SECTION 4

Final Verdict & Recommendation

Dimension	Status
Functional Completeness	READY
Reliability & Error Handling	READY
Security & Access Control	READY
Data Integrity & Multi-Tenancy	READY
AI / LLM Governance	READY
Performance & Scalability	READY — NOTED LIMITATION
Observability & Auditability	READY
Operational Readiness	READY

Recommendation

Data Alchemist is recommended for production use. Seven of eight assessed dimensions are unconditionally ready; the eighth (performance & scalability) is ready today with one documented, low-priority limitation that only becomes relevant at a scale this deployment has not yet reached. No open item in Section 3 is rated as blocking.

This assessment reflects the system's state as verified on 2026-07-09 and should be re-run after material changes to the connector, agent, or data-model layers.

VERDICT: READY FOR PRODUCTION